Chapter -I- Fundamentals of Graph Theory

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Academic year 2026 - 2027

2 octobre 2026

1 Introduction

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2 What is a graph ?

1. Introduction
2. What is a graph ?
3. Some more applications of graphs

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3. Some more applications of graphs
4. Basic concepts of Graph Theory

1. Introduction
2. What is a graph ?
3. Some more applications of graphs
4. Basic concepts of Graph Theory
5. Order, size, adjacency, degree and incidence in a graph

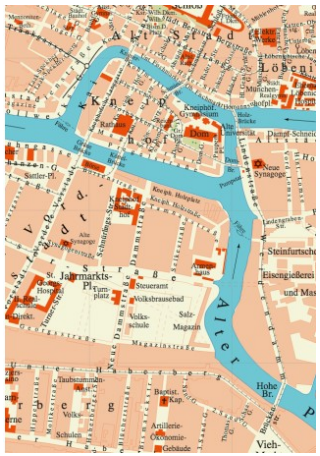
Why Graph Theory ?

- Graphs are used to model pair wise relations between objects.
- Generally, a network can be represented by a graph.
- Many practical problems can be easily represented using graph theory.
- Much of computer science operates through the usage of graph algorithms and data structures.

History of Graph Theory

The Königsberg bridge problem

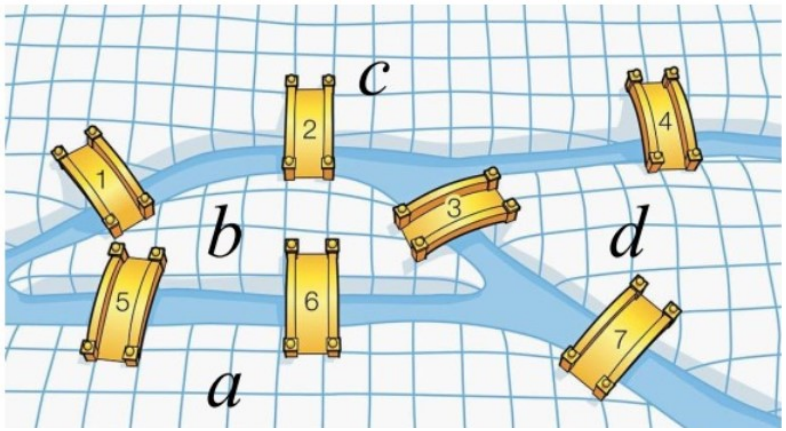
- Königsberg was a city in Prussia (now Kalingrad, Russia)
- The city was situated in both sides of the Pergel River and included 2 large Islands which were connected to each other, or to the mainland of the city, by 7 bridges.



- **PROBLEM :** Can you take a walk through the town, visiting each part of the town and crossing each bridge exactly once ?

History of Graph Theory

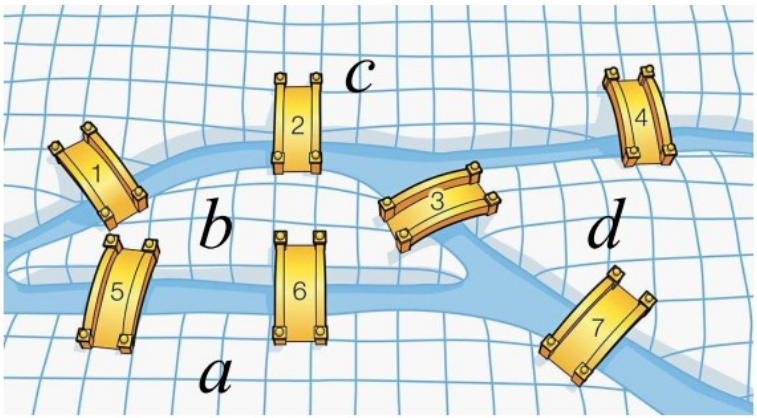
We can simplify the Königsberg's map to :



History of Graph Theory

The Königsberg bridge problem

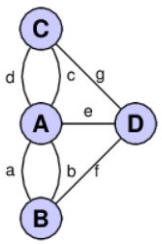
- Königsberg is made up of 4 landmasses and 7 bridges
- People wondered if one could walk across every bridge exactly once
- Leonhard Euler tried to solve it
- He drew out the city in a more simple way



The City of Königsberg

Leonhard Euler





The Königsberg bridge problem

The Königsberg Bridge Problem asks if it's possible to cross all seven bridges of Königsberg exactly once without repeating any bridge. This was one of the earliest problems in graph theory, solved by Euler.

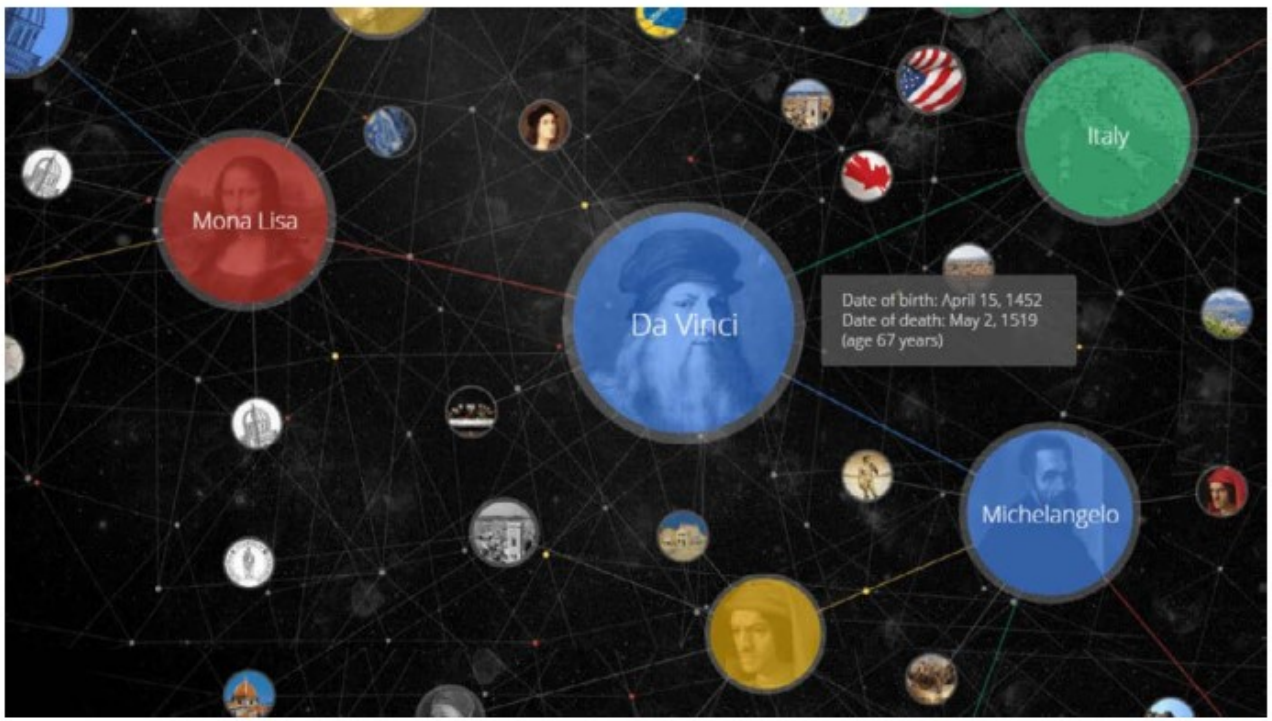
- The landmasses are treated as vertices and the bridges as edges.
- Eulerian paths require exactly two vertices of odd degree (number of edges).
- In the Königsberg problem, all four vertices have an odd degree.

Conclusion : *There is no solution to the Königsberg Bridge Problem because it violates the conditions for an Eulerian path.*

What is a graph ?

- Vertices and edges can have types and metadata

Google Knowledge Graph

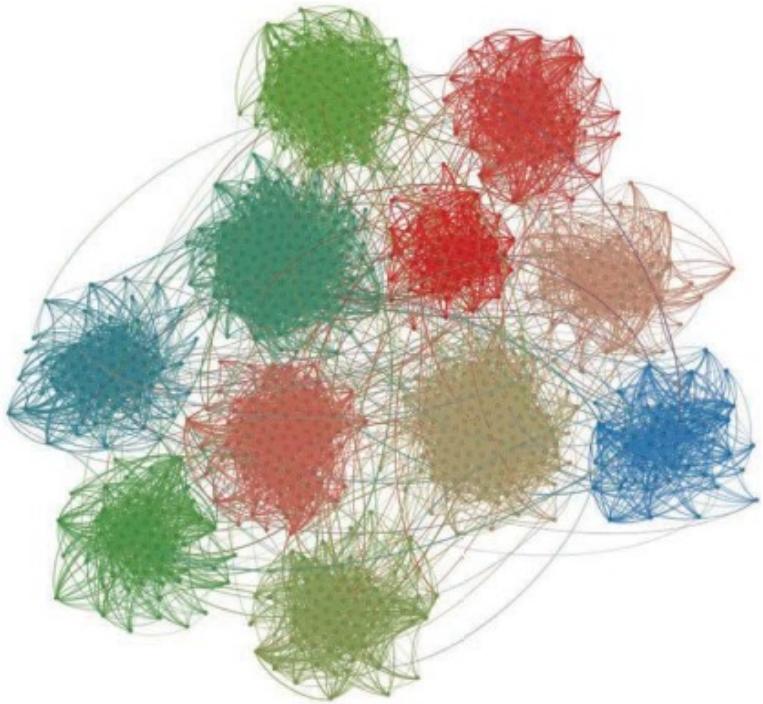


Social network queries

Examples

1. Finding all your friends who went to the same high school as you
2. Finding common friends with someone
3. Social networks recommending people whom you might know
4. Product recommendation

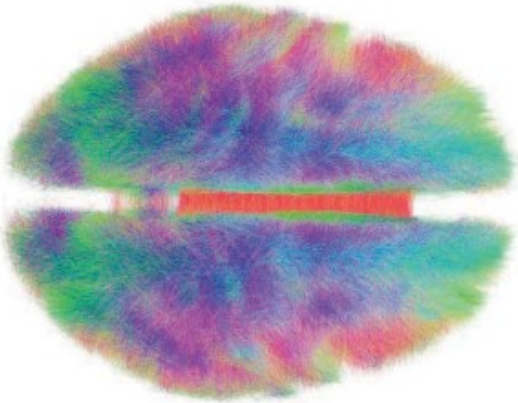
Finding good clusters



- Some applications
  - Finding people with similar interests
  - Detecting fraudulent websites
  - Document clustering
  - Unsupervised learning

- Finding groups of vertices that are "well–connected" internally and "poorly–connected" externally

More Applications



Connectomics

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- Study of the brain network structure

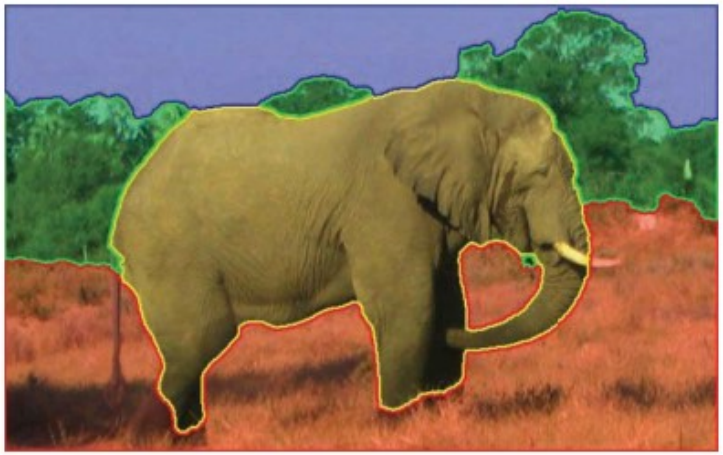


Image Segmentation

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- Pixels correspond to vertices
- Edges between neighboring pixels with weight corresponding to similarity

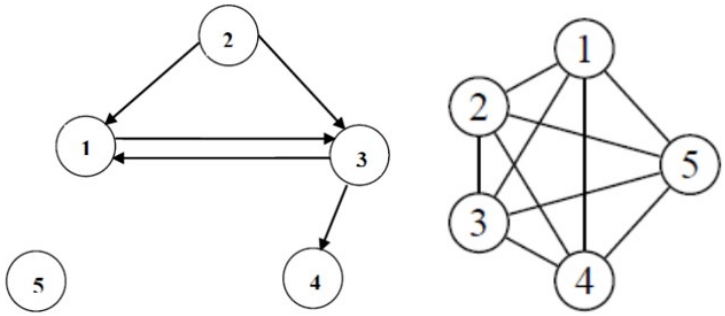
Definition : Graph

Graphs allow you to visualize and manipulate relationships between objects through a graphical representation. **Objects** are represented by **points** called **vertices or nodes**. **Relationships** are represented by **branches**.

1. A graph is a collection of nodes and edges
2. Denoted by $G = (V, E)$
   - $V =$ **nodes**(vertices, points)
   - $E =$ **edges**(links, arcs) between pairs of nodes.
   - Graph size parameters : n $= |V|$, m $= |E|$.

Definition : Graph

1. If the branches are **oriented**, they are called **arcs** and the graph is said to be **directed**.
2. If the branches are **non-oriented**, they are called **edges** and the graph is said to be **undirected**.



Directed Graph Undirected Graph

3. If the branches are **weighted**, the graph is said to be **weighted**.

Definition : Undirected Graph

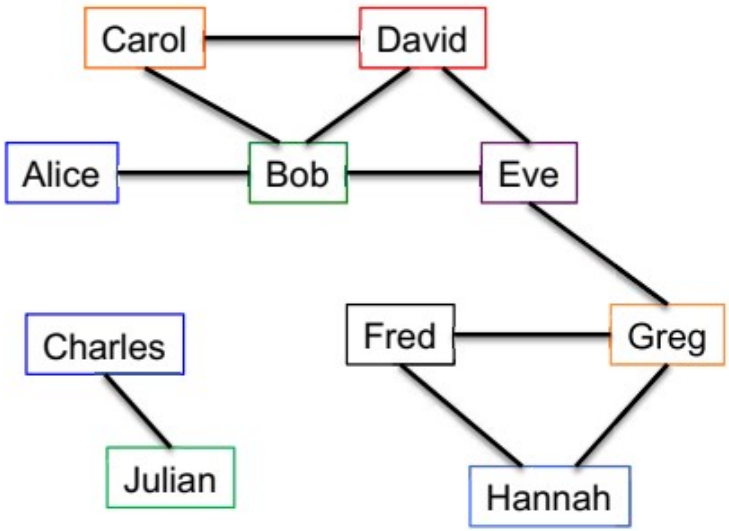
A finite **undirected** graph $G = (V, E, \Psi)$ is defined by :

- the finite set V whose elements are called **vertices**,
- the finite set E whose elements are called **edges**,
- the function $\Psi : E \to V \times V$ called incidence function which associates with each edge $u \in E$ a **non-ordered pair** $\{I(u), T(u)\}$ of vertices of V called the **endpoints** of u.
    - If I(u) = a and T(u) = b, we will say that the edge u connects the vertices a and b, that the vertices a and b are adjacent or incident with u or, again, that the edge u is incident with a and b.
    - An edge $u = (a, b) \in E$ is symmetric $\Leftrightarrow (b, a) \in E$

Undirected Graph



- Vertices model objects
- Edges model relationships between objects



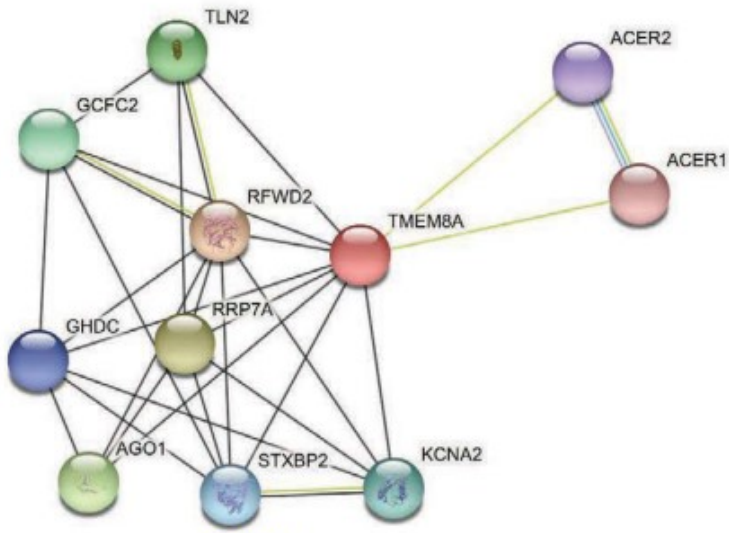
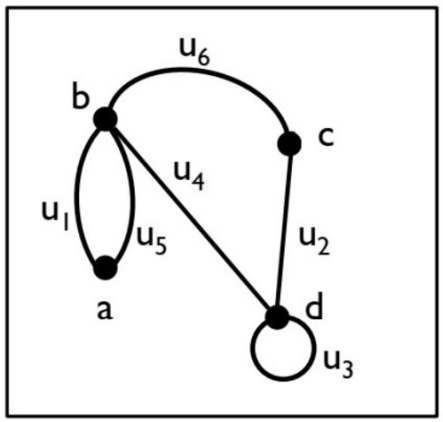


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Graphical representation

Undirected Graph

We associate a point with each **vertex** and we represent each edge by a simple curve connecting the two points associated at its ends.



$V = \{a, b, c, d\}$
$E = \{u_1, u_2, u_3, u_4, u_5, u_6\}$

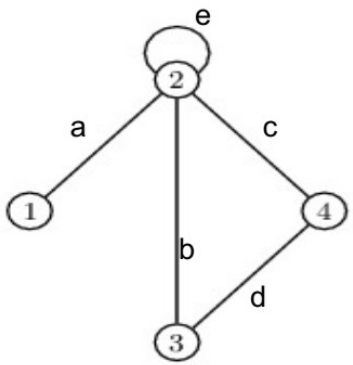
ψ	u_1	u_2	u_3	u_4	u_5	u_6
I(u)	a	d	d	b	a	b
T(u)	b	c	d	d	b	c

Remark : A graph has an infinite number of representations.

Definition : Undirected Graph

Example

- The set of vertices : $V = \{1, 2, 3, 4\}$
- The set of edges : $E = \{a, b, c, d, e\}$
- $a = \{1, 2\}; b = \{2, 3\}; c = \{2, 4\}; \ldots; d = \{2, 2\}$
- e is a **loop**.

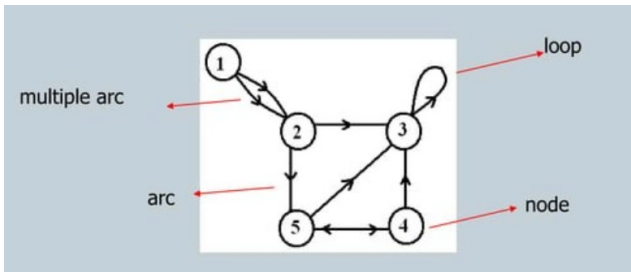


	a	b	c	d	e
I(e)	1	2	2	3	2
T(e)	2	3	4	4	

Definition : Directed Graph

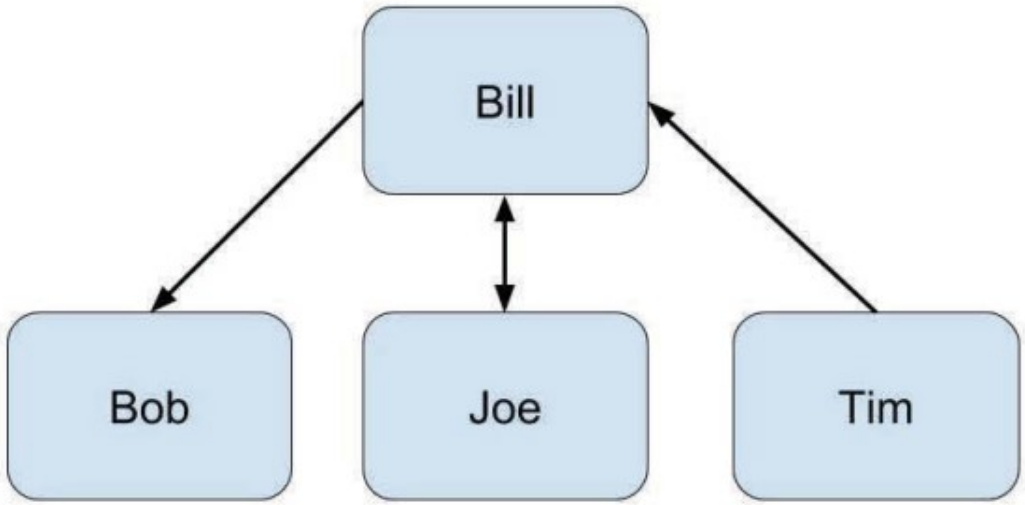
A finite **directed** graph $G = (V, E, \Psi)$ is defined by :

- the finite set V whose elements are called **vertices**,
- the finite set E whose elements are called **arcs**,
- the function $\Psi : E \to V \times V$ called incidence function which associates with each edge $u \in E$ a **ordered pair** $\{I(u), T(u)\}$ of vertices of V called the **endpoints** of u.
  - I(u) is called a **source** (head, origin).
  - T(u) is called a **target** (tail, destination).



Directed Graph

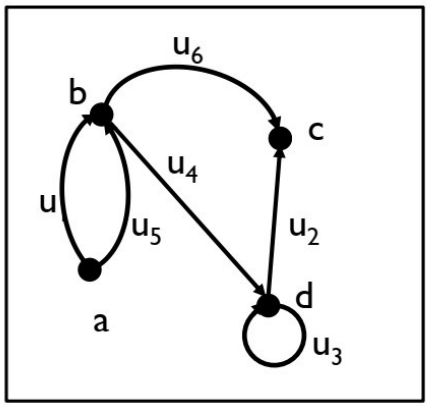
- Edges can be directed
  - Relationship can go one way or both ways



Graphical representation

Directed Graph

We associate a point with each **arc** and we represent each edge by a simple curve connecting the two points associated at its endpoints. In order to identify the direction of each arc, we add a directed arrow towards the arc's target.

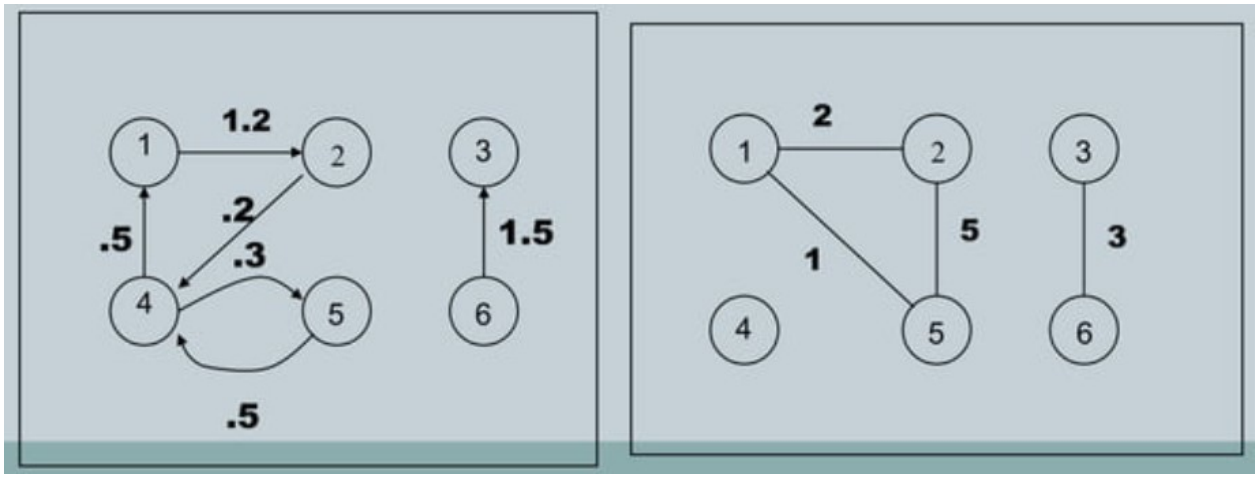


$V = \{a, b, c, d\}$
$E = \{u_1, u_2, u_3, u_4, u_5, u_6\}$

ψ	u_1	u_2	u_3	u_4	u_5	u_6
I(u)	a	d	d	b	a	b
T(u)	b	c	d	d	b	c

Weighted graph

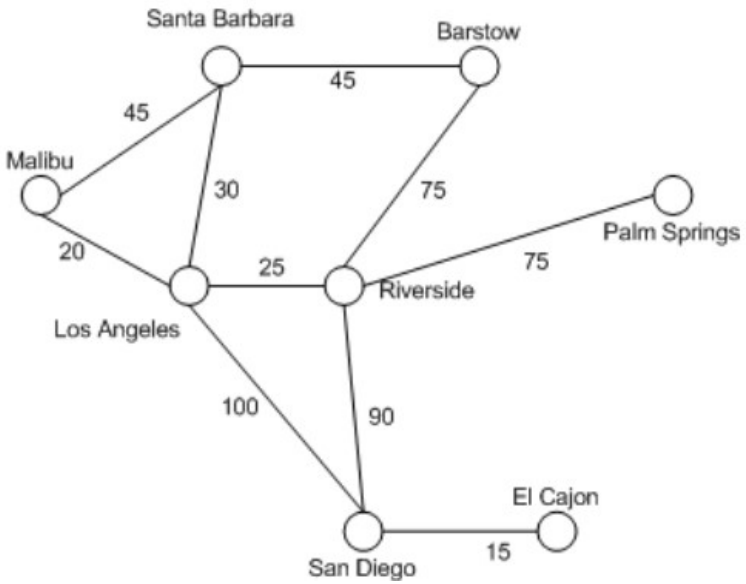
A weighted graph is a graph for which each edge has an associated weight.



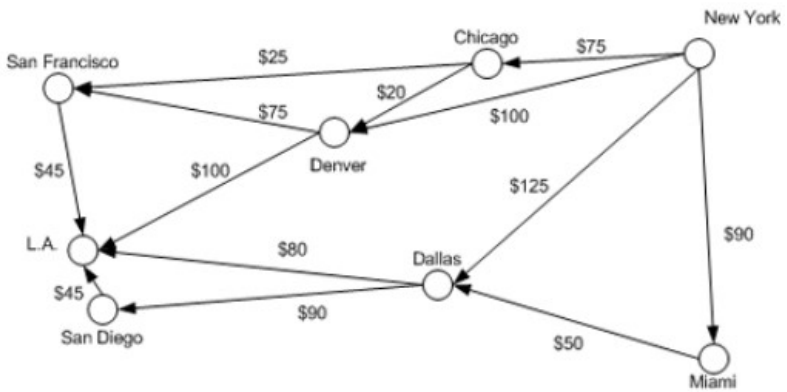
Weighted graph

- Edges can be weighted
  - Denotes “strength”, distance, etc.

Distance between cities



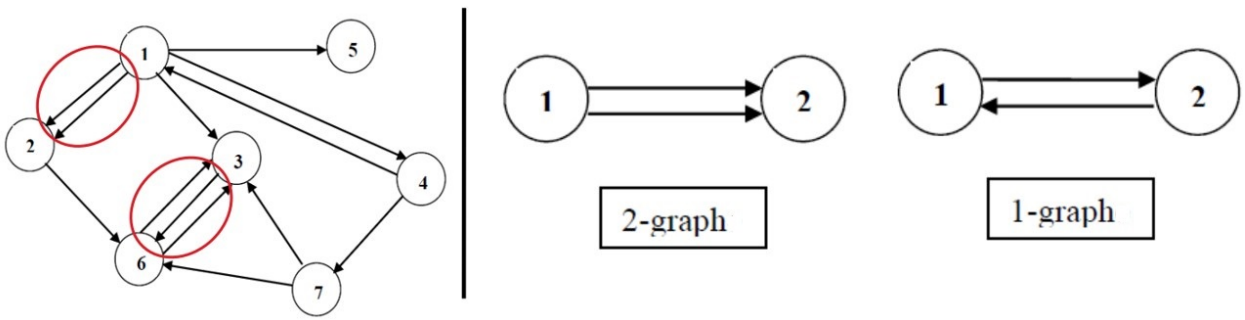
Flight costs



Fundamental concepts

p-graph

1. A graph is called a p-graph if at most p arcs go from any vertex to any other vertex.
2. In other words, a p-graph is a graph in which there are no more than p arcs between two vertices.
3. Example



Terminology

1. Adjacent vertices : directly linked by an edge.
2. Adjacent edges : share a common end vertex.
3. An edge is **incident** to a vertex if it connects that vertex to another vertex.

"Multivoque" applications

A 1-graph G can be defined by a finite set of vertices V and a "multivoque" application :

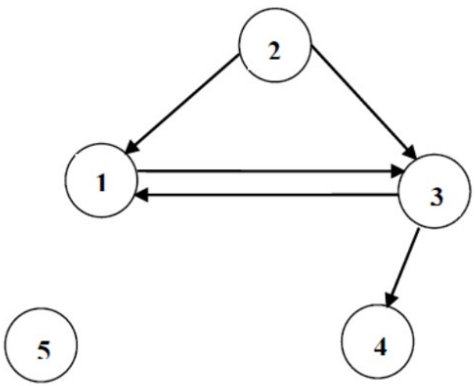
$$\begin{array}{ccccc} \Gamma & : & V & \rightarrow & V \\ & & x & \mapsto & \Gamma(x) \end{array}$$

Where $\Gamma(x) = \{y/(x,y) \in E\}$ (E is the set of edges).

- The set of successors of a vertex x is denoted $\Gamma(x)$
- The set of predecessors of a vertex is denoted $\Gamma^{-1}(y)$

"Multivoque" applications

Example

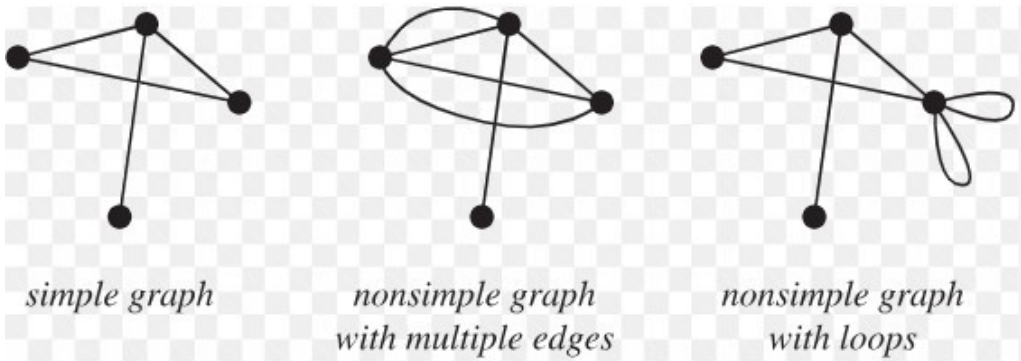


$\Gamma(1)=\{3\}$
$\Gamma(2)=\{3, 1\}$
$\Gamma(3)=\{1, 4\}$
$\Gamma(4)=\{\emptyset\}$
$\Gamma(5)=\{\emptyset\}$

$\Gamma^{-1}(1)=\{2, 3\}$
$\Gamma^{-1}(2)=\{\emptyset\}$
$\Gamma^{-1}(3)=\{1, 2\}$
$\Gamma^{-1}(4)=\{3\}$
$\Gamma^{-1}(5)=\{\emptyset\}$

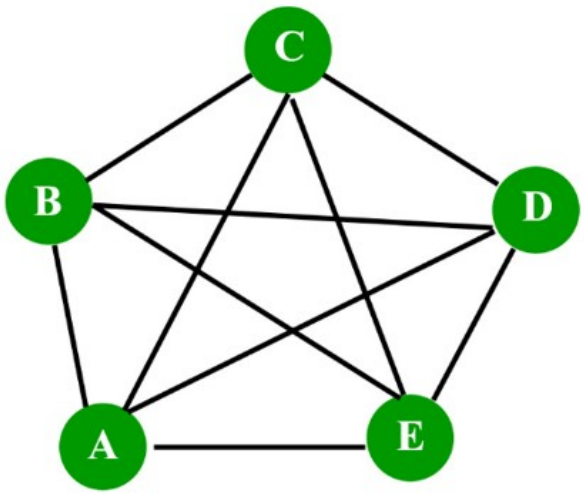
Simple graph

- A graph is said to be **simple** if it has neither loop nor two identical edges (arcs).
- Example :



Complete graph

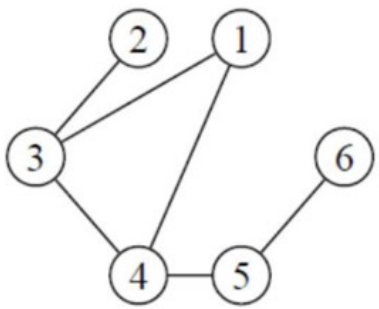
- A graph is said to be **complete** if each vertex of the graph is directly connected to all the other vertices.
- Example :



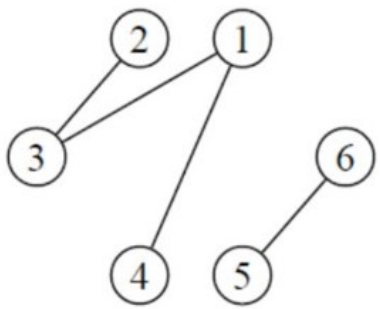
Complete Graph

Partial graph

- Let $G = (V, E)$ be a graph, and $E' \subset E$, then $G' = (V, E')$ is a **partial** graph of G. In other words, we obtain G' by removing one or more edges from the graph G.
- Example :



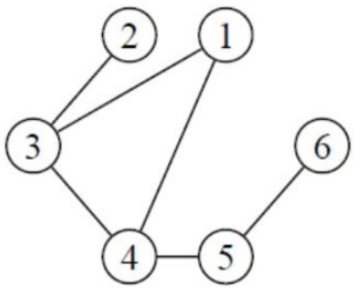
Graph G



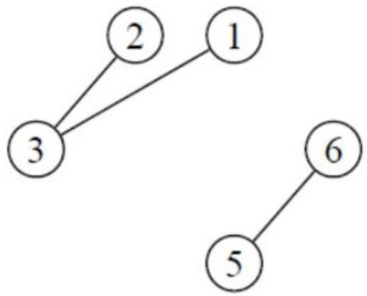
Partial graph of G

Sub-graph

- Let $G = (V, E)$ be a graph, $V' \subset V$, and $E' = \{(x, y) \in E / x \in V' \text{ and } \quad y \in V'\}$, then, $G'(V', E')$ is called a sub-graph of G. In other words, we obtain G' by removing some vertices from the graph G, but, we keep all the arcs which connect the remaining vertices.
- Example :



Graph G



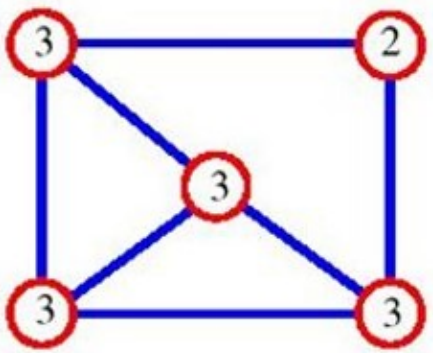
Sub-graph of G

Order, size and adjacency

1. We call **order** of a graph the number of its vertices, $|V| = n$.
2. We call **size** of a graph the number of its edges (arcs), $|E| = m$.
3. Two **vertices** connected by an edge (an arc) are said to be **adjacent**.
4. Two **edges (arcs)** having a common target are said to be **adjacent**.

Degree and Incidence in Undirected Graphs

- The **degree** of a vertex is the number of edges that are incident to that vertex.



Degree and Incidence in Directed Graphs

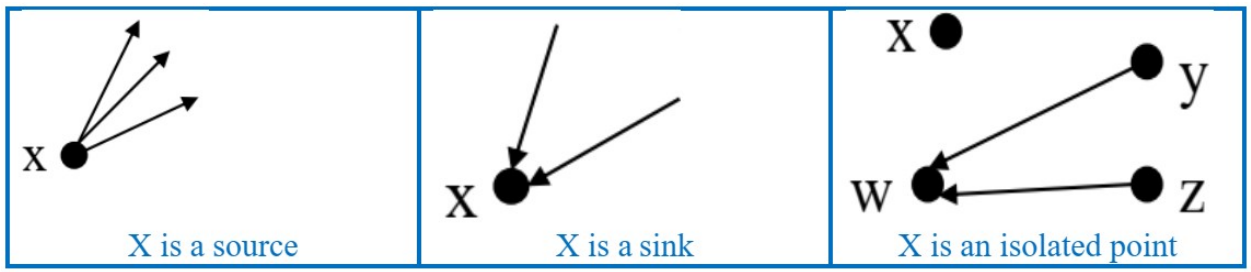
- Let $u = (x, y) \in E / x \neq y$
-



  - We say that the arc u is **entering** to y.
  - We say that the arc u is **leaving** x.
- The In-degree $d^-(x)$ of vertex x is the number of arcs **entering** to x.
- The Out-degree $d^+(x)$ of vertex x is the number of arcs **leaving** x.

Degree and Incidence in Directed Graphs

1. The degree of x is defined by : $d(x) = d^-(x) + d^+(x)$.
   The degree of a vertex is edges that are incident to that vertex.
2. x is called an input or **source** if $d^-(x) = 0$ and $d^+(x) > 0$.
3. x is called an output or **sink** if $d^+(x) = 0$ and $d^-(x) > 0$.
4. x is an **isolated point** if $d^+(x) = 0$ and $d^-(x) = 0$.
5. **Example :**



Degree and Incidence in Directed Graphs

Properties

Let G be a graph with n vertices and m edges.

Property 1

$$\sum_{v} deg(v) = 2m$$

Proof: each edge is counted twice.

Property 2

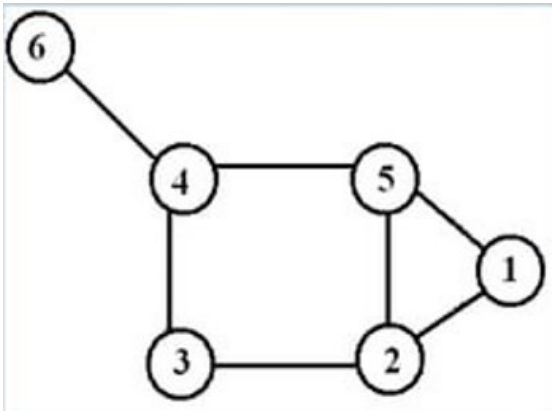
If G is undirected with no self-loops and no multiple edges:

$$m \leq n(n-1)/2$$

Proof: the maximum number of edges is obtained when each vertex is connected to all the other vertices of the graph. We have,

$$(n-1) + (n-2) + (n-3) + \ldots + 2 + 1 = n(n-1)/2$$

Example



1. $G = (V, E)$
2. $V = \{1, 2, 3, 4, 5, 6\}$
3. $E = \{\{1, 2\}; \{1, 5\}; \{2, 3\}; \{2, 5\}; \{3, 4\}; \{4, 5\}; \{4, 6\}\}$
4. $|V| = 6$, $|E| = 7$.
5. $N(4) = \text{Neighborhood}(4) = \{3, 5, 6\}$

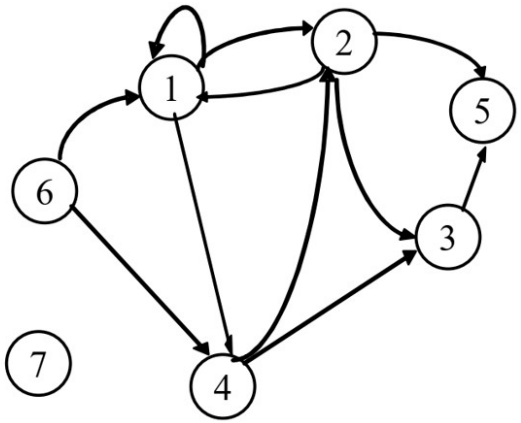
Exercise

Consider the directed graph $G = (V, E, \Psi)$ such that :
$V = \{1, 2, 3, 4, 5\}$
$E = \{(1, 2); (1, 4); (2, 2); (2, 3); (2, 4); (3, 5); (4, 3); (5, 3)\}$

1. Draw the diagram of graph G.
2. Represent in a table the incidence function Ψ.
3. Give the Out-degree of 2 and the In-degree of 4.
4. Give the set of predecessors of 4 and the set of successors of 2.
5. Give a partial graph and a sub-graph of G.

Application 1

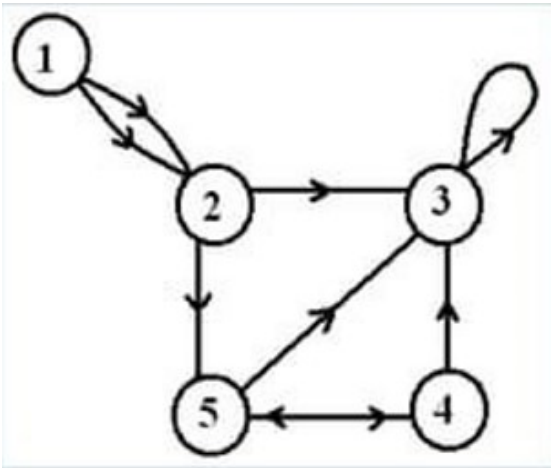
Let us consider the following graph G :



1. Give the order and the size of the graph G.
2. Specify the vertices adjacent to vertex 3.
3. Determine the In-degrees and Out-degrees for each vertex.
4. Determine the vertices which are a source, a sink or an isolated point.

Application 2

Let us consider the following graph G :



1. Give the order and the size of the graph G.
2. Specify the vertices adjacent to vertex 5.
3. Determine the In-degrees and Out-degrees for each vertex.
4. Determine the vertices which are a source, a sink or an isolated point.

References

- *Wilson R.J., (2010), Introduction to Graph Theory, Fifth edition, Pearson*
- *Diestel R., (2017), Graph Theory, Fifth edition, Springer Nature*
- *Balakrishnan V.K., (1997), Theory and problems of Graph Theory, Shhaum's outline series, McGraw-HILL.*
- *Introduction to Graph Theory, University of California San Diego, (20 hours), COURSERA.*